

Mini SIEM Capstone Project – Week 7 Progress Report

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Project Overview

The goal of this project is to design and implement a simplified Security Information and Event Management (SIEM) system capable of ingesting log data, normalizing events into a unified schema, storing them in a database, and performing detection and correlation across multiple data sources. The long-term objective is to simulate real-world SIEM functionality, including identifying and validating malicious activity using labeled datasets.

Summary of Week 7 Work

During Week 7, the focus shifted from working with independent datasets to integrating a correlated, multi-source dataset. The previously used datasets (auth.log and honeypot logs) were sufficient for building the ingestion and detection pipeline, but they lacked shared identifiers such as user, host, and timeline, which prevented meaningful correlation.

To address this limitation, the project transitioned to the Los Alamos National Laboratory (LANL) cybersecurity dataset. This dataset provides multiple log sources with shared identifiers, enabling cross-source correlation. For this week, two key data sources were integrated:

- auth.txt (authentication events)
- redteam.txt (labeled attacker activity)

The system was modified to support these new formats while maintaining the existing architecture. New normalization functions were created to map LANL data into the existing NormalizedEvent schema. The ingestion pipeline, database storage, and event processing flow remained unchanged.

Implementation Details

1. LANL Authentication Data Integration

A new parser (lanl_auth.py) was created to process authentication events. Each record was parsed into fields such as user, source computer, destination computer, authentication type, and logon activity. These events were normalized into the existing schema and stored in the SQLite database.

A targeted sample of 20,000 authentication records was extracted from the dataset to focus on the time range where known attack activity occurs.

LANL Authentication Integration Output:

```
PS C:\Users\trent\OneDrive\Desktop\mini_siem> python lanl_main.py
=== LANL Auth Integration Test ===
Input file: data/raw/auth_targeted_sample.txt
Normalized events written to: data/normalized/lanl_auth_events.jsonl
Total events normalized: 20000
Parse failures: 0

Top event types:
  lanl_logoff: 8394
  lanl_logon: 8308
  lanl_auth_other: 3298
```

2. Red Team Data Integration (Ground Truth)

A second parser (lanl_redteam.py) was developed to process labeled attack events from redteam.txt. These events represent confirmed malicious activity and were stored in the same database with a distinct source identifier.

A total of 749 red team events were successfully ingested and normalized.

Red Team Dataset Integration Output:

```
=== LANL Red Team Integration Test ===
Input file: data/raw/redteam.txt
Normalized events written to: data/normalized/lanl_redteam_events.jsonl
Total events normalized: 749
Parse failures: 0

Top event types:
  lanl_redteam_activity: 749
```

3. Database Integration and Verification

Both LANL authentication and red team events were inserted into the existing SQLite database. Verification queries confirmed:

```
('lanl_auth', 'lanl_auth_other', 3298)
('lanl_auth', 'lanl_logoff', 8394)
('lanl_auth', 'lanl_logon', 8308)
('lanl_redteam', 'lanl_redteam_activity', 749)
```

- 20,000 authentication events
- 749 red team events
- Proper classification of event types (logon, logoff, other, redteam activity)

4. Initial Correlation Analysis

A correlation script (lanl_correlation.py) was developed to analyze relationships between authentication events and red team activity. The system compared shared attributes such as source and destination computers across both datasets.

Results of the correlation analysis:

```
=== LANL Correlation Check ===
LANL auth events: 20000
LANL redteam events: 749
Matching source computers: 0
Matching destination computers: 49818
```

Correlation results showing overlap between authentication events and red team target systems.

- Matching source computers: 0
- Matching destination computers: 49,818

This indicates that while attacker source systems were not present in the authentication sample, a significant number of destination systems targeted by the red team activity appeared within authentication logs. This demonstrates that the system can successfully identify overlap between normal activity and known attack targets.

Sample correlation output showed aligned records including:

```
Sample destination computer matches:
('1931895', 'U737@DOM1', 'C19932', 'C612', 'lanl_logon', 'ANONYMOUS LOGON', 'C357', 'C612', '150')
('464545', 'U737@DOM1', 'C19932', 'C612', 'lanl_logon', 'ANONYMOUS LOGON', 'C357', 'C612', '150')
('584564', 'U737@DOM1', 'C19932', 'C612', 'lanl_logon', 'ANONYMOUS LOGON', 'C357', 'C612', '150')
('1079883', 'U66@DOM1', 'C17693', 'C1085', 'lanl_logon', 'C104$', 'C105', 'C1085', '150')
('578438', 'U737@DOM1', 'C19932', 'C1065', 'lanl_logoff', 'C1065$', 'C1065', 'C1065', '150')
('578438', 'U737@DOM1', 'C19932', 'C1065', 'lanl_logon', 'C1065$', 'C1065', 'C1065', '150')
```

- red team user and source/destination systems
- corresponding authentication events
- matching destination computers
- associated timestamps

This confirms that the SIEM is now capable of performing basic multi-source correlation.

Key Takeaways

The most significant outcome of this week is the successful transition from isolated log analysis to correlated event analysis. The system now operates on data that supports real-world SIEM functionality, allowing relationships between events to be identified across multiple sources.

The use of labeled red team data also introduces the ability to validate detection logic against known attack activity, which will be critical in upcoming phases of the project.

Challenges Encountered

Several challenges were encountered during this phase:

- Handling large dataset sizes and creating manageable samples
- File encoding inconsistencies (UTF-8 vs UTF-16) during ingestion
- Ensuring correct mapping of LANL data fields into the existing schema
- Avoiding duplicate database insertions during testing

Each issue was resolved without requiring changes to the overall system architecture.

Next Steps (Week 8 Plan)

The next phase of the project will focus on expanding correlation capabilities and incorporating additional data sources:

- Integrate proc.txt (process execution events)
- Begin building behavior-based correlation (auth → process relationships)
- Develop more advanced detection logic based on event chains
- Continue validating results using red team data

These steps will move the system closer to a fully functional SIEM capable of identifying complex attack patterns.